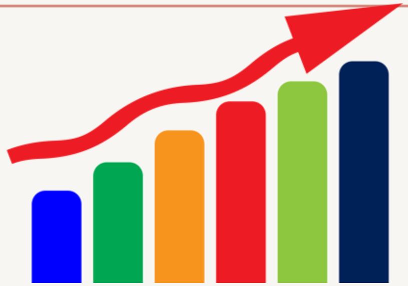
Asma Abbas

Progress Report

Data and cleaning

- The datasets I've primarily used so far were Montgomery County Socioeconomic characteristics, and the MCPS dataset.
- Key variables include: Poverty rate, FARMS rate, graduation rate
- To begin, I did some cleaning for the datasets
    - That included removing any missing values, and doing a general cleaning
    - I also had to rename the variables in my datasets, to make them easier to work with and label.
    - Some of the variables were mislabeled as not numeric, so I had to work on fixing that as well.
    - So far, I've discovered summary statistics about the data, and made general graphs to help answer my questions, and an EDA

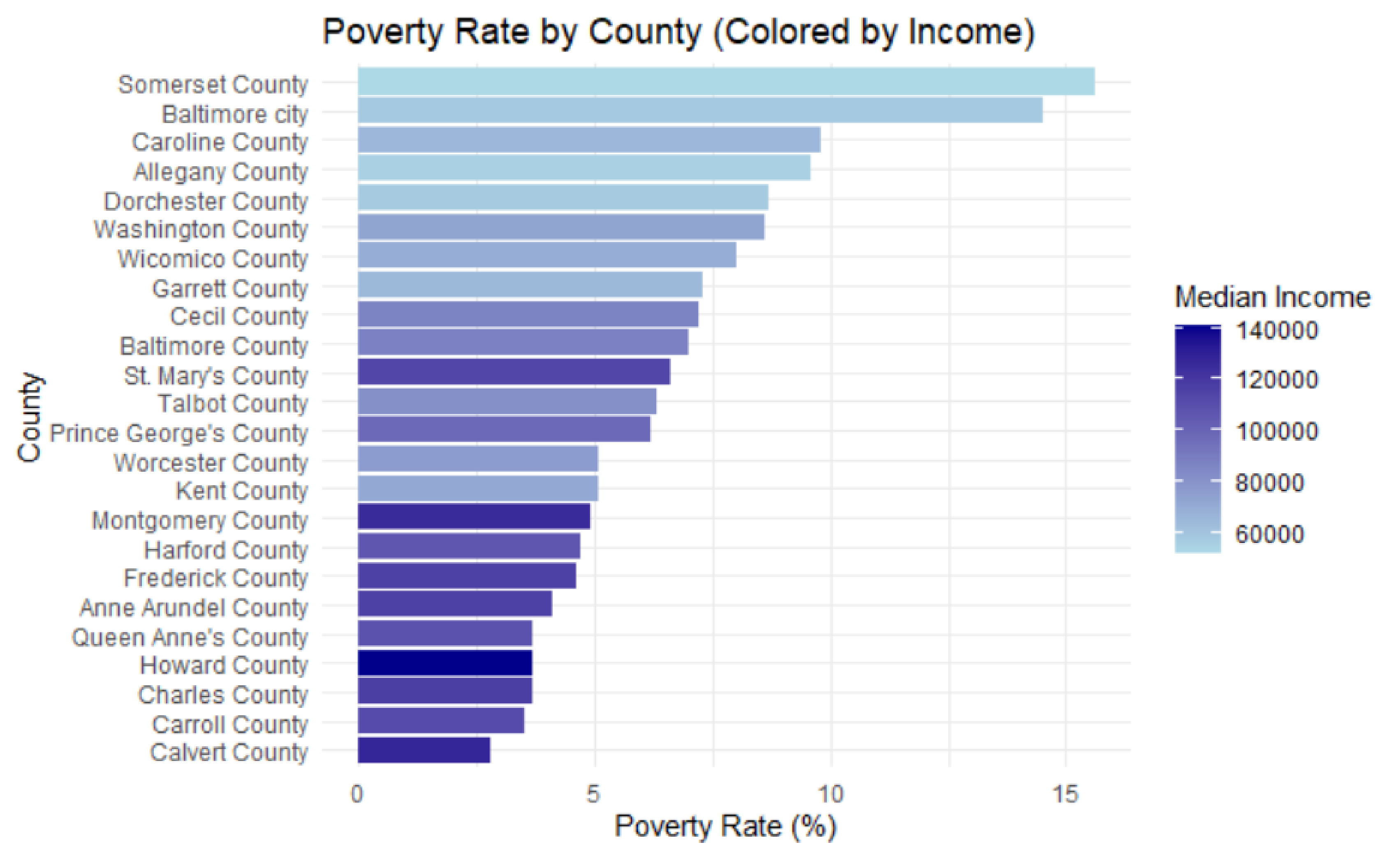
Summary Stats



- From MCPS dataset:
  - Average graduation rate: 87.91
  - Median graduation rate: 89.55
  - The average amount of FARMS rate: 39.16
  - Median farms: 44.8

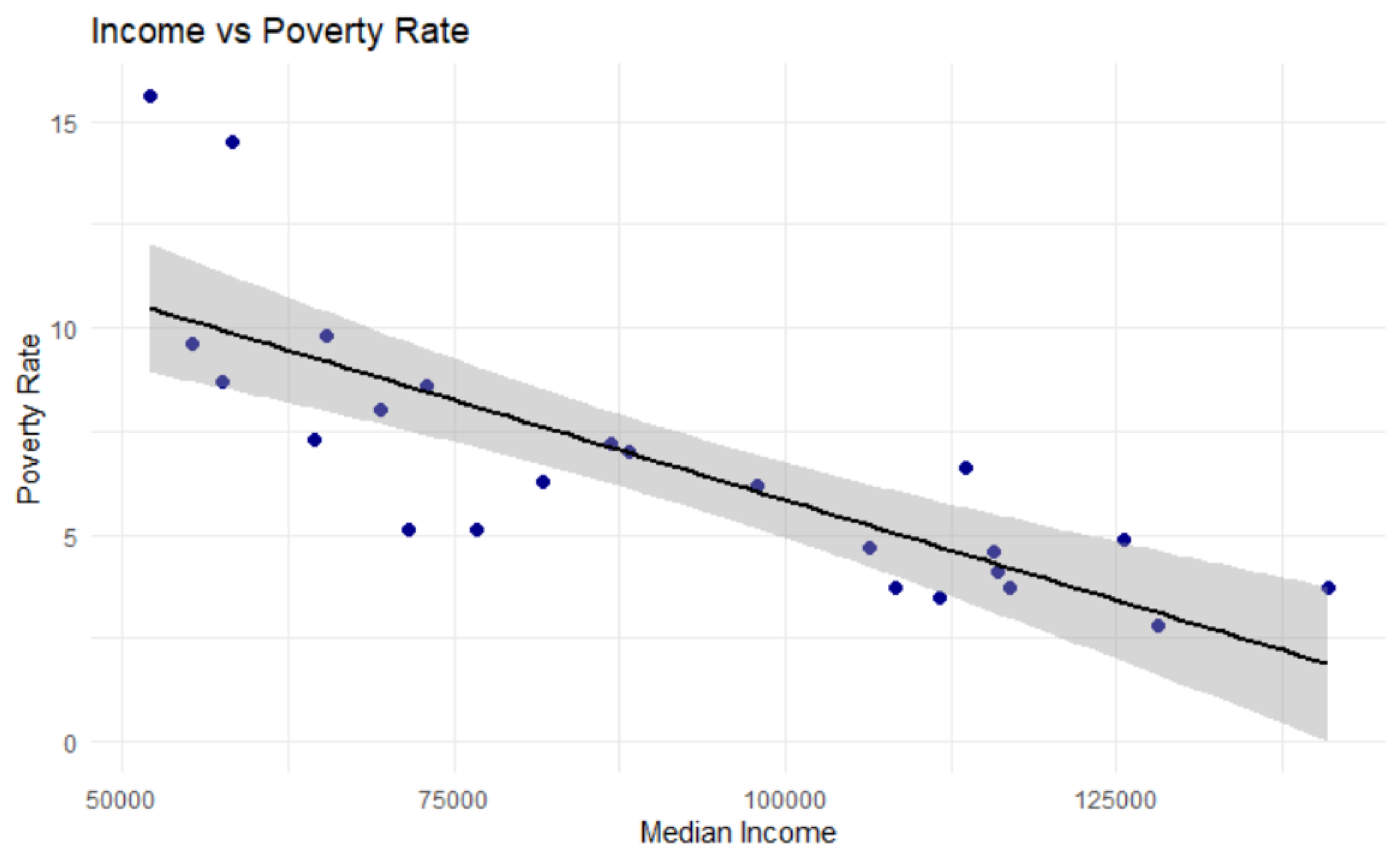
- From Maryland Counties dataset:
  - Average poverty rate is around 6.72%
  - Range: 2.8% to 15.6%
  - Average unemployment rate: 4.9%
  - Average median income: 90,906

Graph 1:

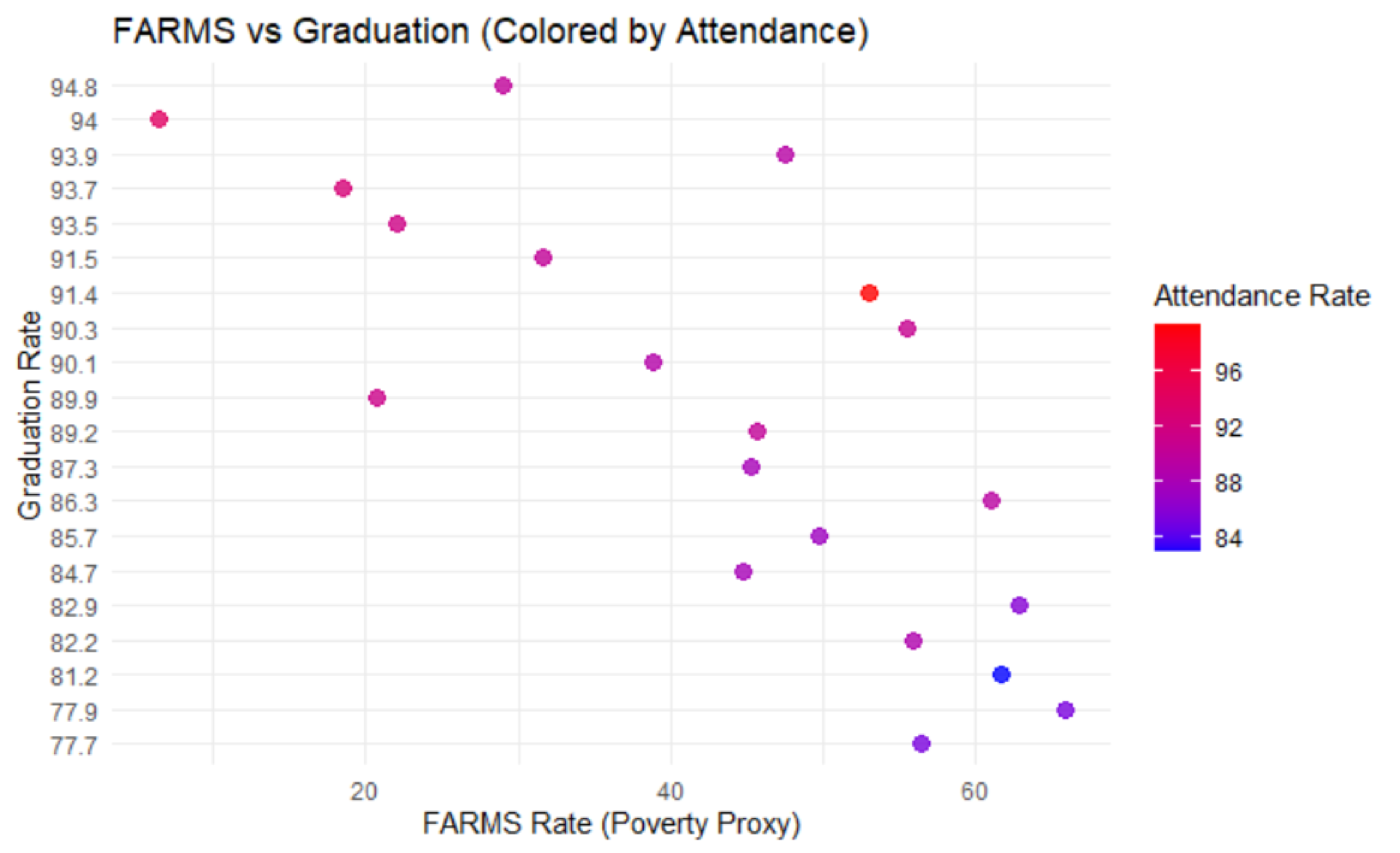


Poverty Rate by County (Colored by Income)
Somerset County
Baltimore city
Caroline County
Allegany County
Dorchester County
Washington County
Wicomico County
Garrett County
Cecil County
Baltimore County
St. Mary's County
Talbot County
Prince George's County
Worcester County
Kent County
Montgomery County
Harford County
Frederick County
Anne Arundel County
Queen Anne's County
Howard County
Charles County
Carroll County
Calvert County
County
0
5
10
15
Poverty Rate (%)
Median Income
140000
120000
100000
80000
60000

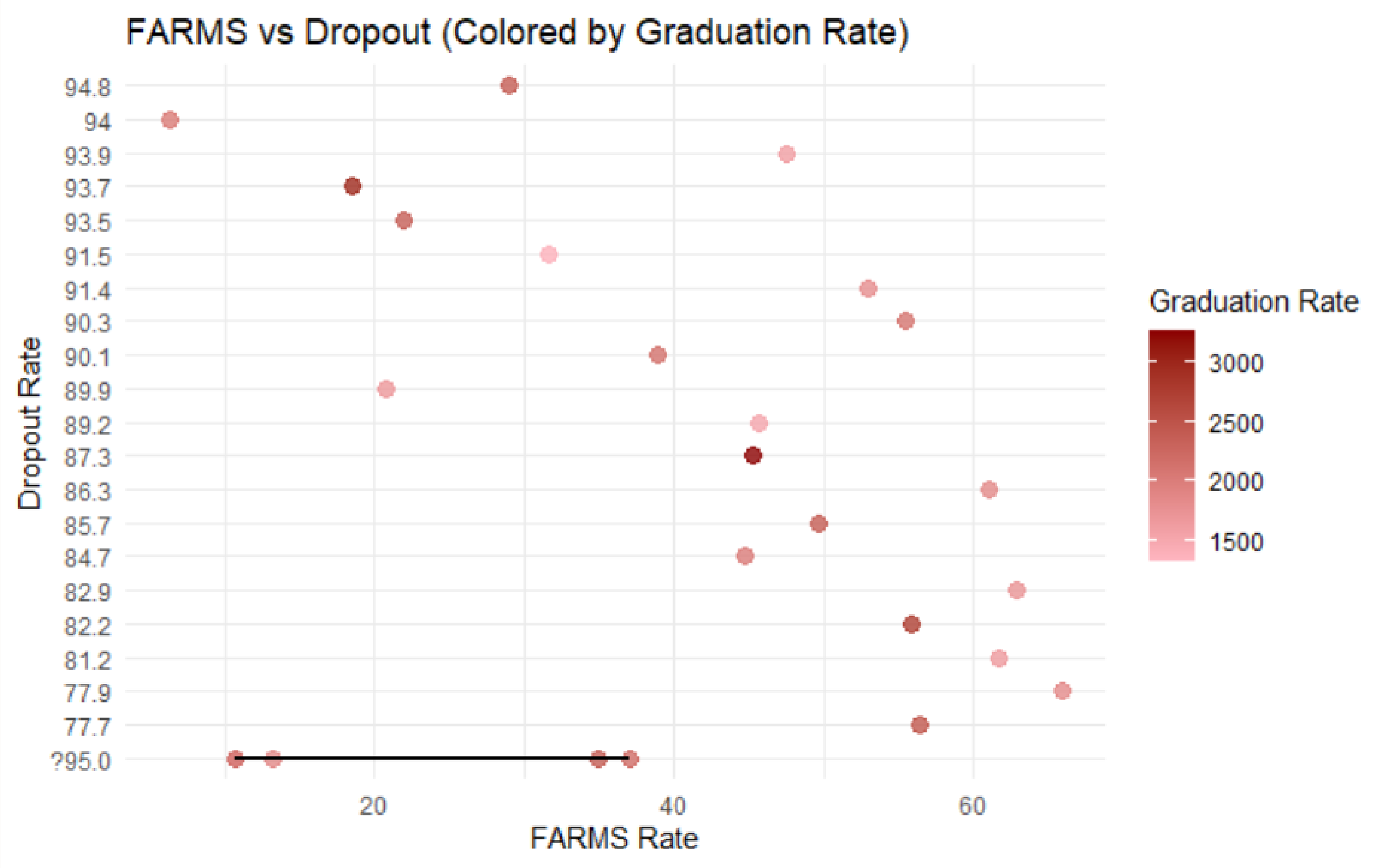
Graph 2


Income vs Poverty Rate
Poverty Rate
15
10
5
0
50000
75000
100000
125000
Median Income

Graph 3: Poverty indicators and their association with graduation rates



Graph 4


FARMS vs Dropout (Colored by Graduation Rate)
Dropout Rate
94.8
94
93.9
93.7
93.5
91.5
91.4
90.3
90.1
89.9
89.2
87.3
86.3
85.7
84.7
82.9
82.2
81.2
77.9
77.7
?95.0
20
40
60
FARMS Rate
Graduation Rate
3000
2500
2000
1500

Moving forward

- Conduct multiple linear regression, with variables like
  - Poverty
  - Attendance
  - Demographics
- Also attempt to merge datasets for further analysis
- Explore geographic patterns through heatmaps
- Build predictive models for the graduation rates
- Investigate the outliers, and underperforming schools.

Thank You!